

One aspect that can feel intimidating about Fourier analysis is the existence of several variants of the transform. The goal of this text is to give an overview of these different methods, see Figure 1.

Take-away 1: The good news is that the core properties of the Fourier transform carry over to all variants. For instance, linearity, convolution, symmetry relations, and the connection between time- and frequency-domain representations remain valid regardless of whether we use the continuous Fourier transform, the DTFT, the DFT, or their numerical implementations.

Take-away 2: In practical digital signal processing, the FFT is the algorithm we rely on almost exclusively. The FFT (and its inverse) is what we use to convert discrete data between the time and frequency domains. Beyond the useful mathematical properties of the Fourier transform, the ingenuity and computational efficiency of the FFT are what make modern applications possible. Streaming, communication systems, medical imaging, and countless real-time processing tasks would simply not be feasible without the FFT.

Fourier Transform (FT)

In this week's lecture, we extended the concept of the Fourier series to the Fourier transform. Similar to the Fourier series, the Fourier transform is a mathematical tool that allows us to represent a function in terms of its frequency components. However, in contrast to the Fourier series, the Fourier transform results in a **continuous function in frequency** and can be applied to a much broader class of **aperiodic** signals $x(t)$.

The Fourier transform ($x(t) \mapsto X(\omega)$) and its inverse ($X(\omega) \mapsto x(t)$) are defined as:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt, \quad (1)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega. \quad (2)$$

Here, $x(t)$ is the input signal, $X(\omega)$ its Fourier transform, and ω the angular frequency.

Even though these formulas may seem a bit heavy at first, the underlying idea is quite simple: Eq. 1 integrates $x(t)$ over all time (from $-\infty$ to $+\infty$), with the exponential term parametrized by ω , producing the function $X(\omega)$ that describes the signal's frequency content. Eq. 2 performs the inverse operation and reconstructs $x(t)$ from its spectrum.

These integrals may be hard to solve analytically, but can be approximated numerically for nearly any practical signal.

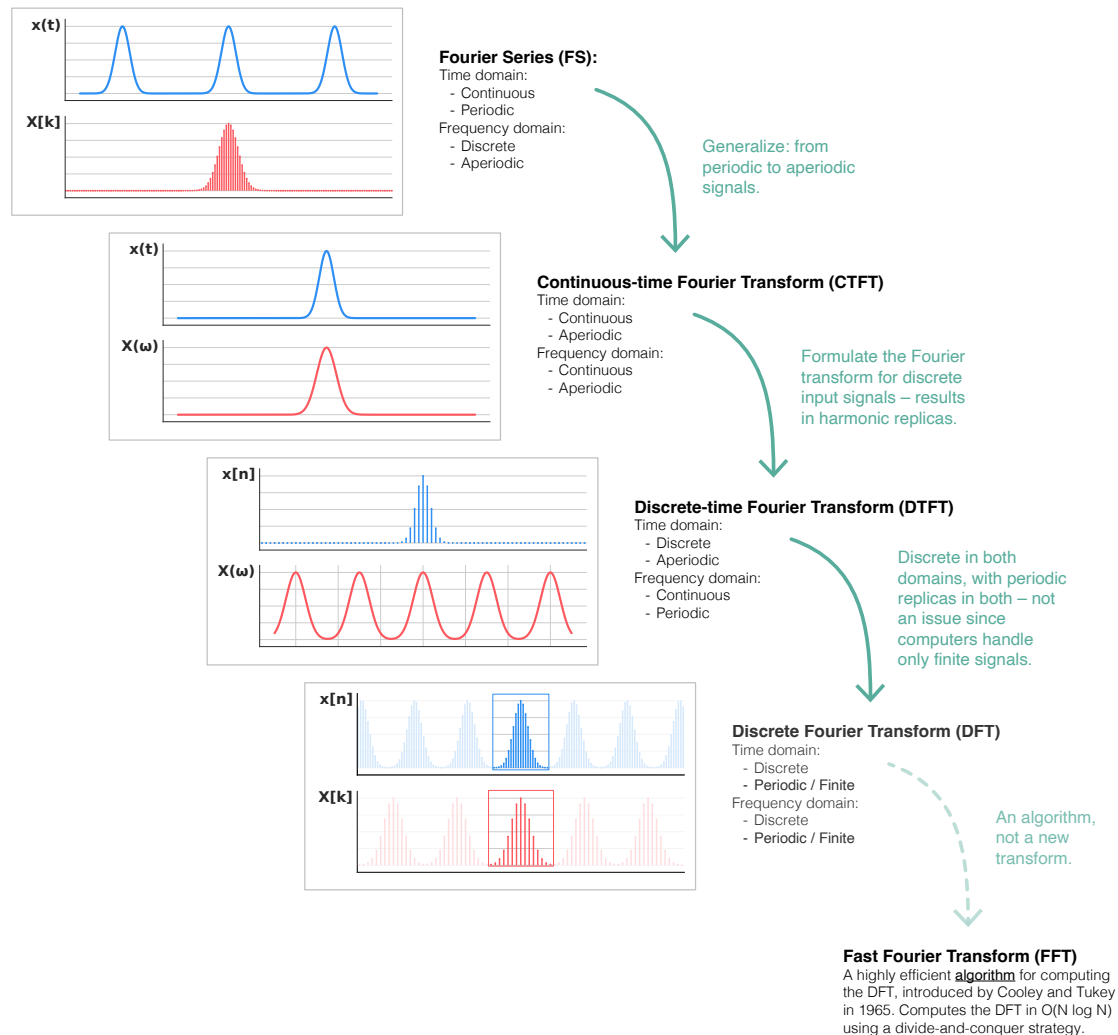


Figure 1: Overview of the Fourier landscape, showing the main transform variants and their approximate historical order – from the Fourier series (1807) to the DFT/FFT (1965). With the Fourier Series and the Fourier Transform, we develop the core intuition that arbitrary signals can be constructed from suitable sinusoidal basis functions. The DTFT and DFT follow naturally when we restrict ourselves to finite, discrete-time signals and therefore require corresponding discrete versions.

Note that using sinusoids as basis functions is not the only possible choice. Other families of basis functions lead to alternative frameworks—[wavelets](#) being a prominent example. But that is a topic for another day.

Discrete Fourier Transform (DFT)

So far, we focused on the **continuous-time** Fourier transform. In practice, however, we typically work with **discrete-time** signals. The discrete-time Fourier transform (DTFT) is the counterpart of the continuous-time transform for such signals: it takes a discrete-time input $x[n]$, but it still produces a **continuous** frequency-domain representation. This is inconvenient for computers, which require **discrete** and **finite** representations.

To fully meet our needs, we require discrete representations in both time and frequency domains. This leads us to the **Discrete Fourier Transform (DFT)**, which provides:

- a finite-length, discrete-time input and
- a finite-length, discrete-frequency output.

The DFT ($x[n] \mapsto X[k]$) and its inverse ($X[k] \mapsto x[n]$) are defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-i2\pi \frac{n}{N}k}, \quad (3)$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{i2\pi \frac{n}{N}k}. \quad (4)$$

Here, $x[n]$ is the discrete-time input signal, $X[k]$ its DFT, and N the number of samples. Note that evaluating Eq. 3 for a fixed value of k —that is, for one discrete frequency—requires computing only a finite sum over the N available samples, and vice versa for the inverse DFT in Eq. 4.

The DFT allows us to analyze the frequency content of discrete-time signals with a **finite number of samples**. We can think of the DFT as a sampled version of the continuous-time Fourier transform. Although its derivation is somewhat complex—requiring sampling in both the time and frequency domains—the DFT retains many key properties of the continuous-time Fourier transform. Specifically, the DFT is linear, satisfies the convolution theorem, and preserves the symmetry properties of the Fourier transform.

Fast Fourier Transform (FFT)

The **Fast Fourier Transform (FFT)** is not a new form of Fourier transform; it is a highly efficient algorithm for computing the DFT $X[k]$ of a discrete signal $x[n]$ with a finite number of samples. Conversely, the **inverse FFT (IFFT)** reconstructs the original signal (or a close approximation) from its frequency-domain representation.

Technical Note: Different FFT implementations may compute the DFT up to a constant factor. SciPy’s default FFT implementation follows the standard DFT convention, where the FFT output scales with the number of samples. To compute the correct magnitude spectrum, we typically divide the FFT result by the number of samples.

Method	Input	Output	Forward Transform	Inverse Transform
FS	Continuous-time periodic signal $x(t)$ with period T	Discrete set of frequency coefficients c_k	$c_k = \frac{1}{T} \int_T x(t) e^{-ik\omega_0 t} dt$, with $\omega_0 = \frac{2\pi}{T}$	$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{ik\omega_0 t}$
CTFT	Continuous-time aperiodic signal $x(t)$	Continuous spectrum $X(\omega)$	$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt$	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega$
DTFT	Discrete-time sequence $x[n]$	Continuous (2π -periodic) spectrum $X(\omega)$	$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-i\omega n}$	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{i\omega n} d\omega$
DFT	Finite discrete sequence $x[n], n = 0, \dots, N-1$	Discrete spectrum $X[k]$	$X[k] = \sum_{n=0}^{N-1} x[n] e^{-i2\pi kn/N}$	$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{i2\pi kn/N}$